(bars: lengths by R, gain by a_{nom}; hats on estimates; $e_\theta = \theta - \hat{\theta}$; posterior quantities unless marked $^-$; $\bar{w}_d[k]$ commanded height, $\Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $\bar{a}' := \bar{a}'_{\mathrm{true}}(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k])$, $\bar{a}'' := \bar{a}''_{\mathrm{true}}(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k])$, both exogenous; $\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta\hat{\bar{w}}[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\hat{\bar{w}}_3[k] + y_1[k] - \delta\hat{\bar{w}}_1[k]\big]$; $e_{\Delta\bar{w}} := \Delta\bar{w} - \Delta\hat{\bar{w}}$; $\mathbb{E}[\cdot]$ mean over noise realizations conditional on the data up to k; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\mathrm{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] + \tfrac{1}{2}\,\bar{a}''_{\mathrm{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k]^2 + \mathcal{O}\big(\Delta\bar{w}^3\big) \\
\bar{a}'_{\mathrm{true}}\big(\bar{w}[k]\big) &= \bar{a}'_{\mathrm{true}}\big(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] - e_{\delta\bar{w}_3}[k]\big) = \bar{a}' - \bar{a}''\,e_{\delta\bar{w}_3}[k] + \mathcal{O}\big(e^2\big) \\
\bar{a}''_{\mathrm{true}}\big(\bar{w}[k]\big) &= \bar{a}'' + \mathcal{O}\big(e\big)
\end{aligned}
$$

$$
\bar{a}_w[k+1] - \bar{a}_w[k] = \big(\bar{a}' - \bar{a}''\,e_{\delta\bar{w}_3}[k]\big)\,\Delta\bar{w}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\bar{w}[k]^2
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k]
\end{aligned}
$$

$$
\begin{aligned}
&\mathbb{E}\big[\varepsilon_w[k]\,n_w[k]\big] = (1-\lambda_c)\,R_1 \\
&\qquad \tilde{\varepsilon}_w[k] := \varepsilon_w[k] - (1-\lambda_c)\,n_w[k], \qquad \mathbb{E}\big[\tilde{\varepsilon}_w\,n_w\big] = 0, \qquad \mathbb{E}\big[\tilde{\varepsilon}_w^2\big] = Q_{33} - (1-\lambda_c)^2 R_1 =: \tilde{Q}_{33} \\
&\qquad \Delta\bar{w}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\bar{w}_3[k] + n_w[k]\big] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}
$$

2 Estimator

$$
\begin{aligned}
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] - (1-\lambda_c)\big[y_1[k] - \delta\hat{\bar{w}}_1[k]\big] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\big[\delta\hat{\bar{w}}_3[k] + y_1[k] - \delta\hat{\bar{w}}_1[k]\big] \\
\hat{\bar{a}}_w[k+1] - \hat{\bar{a}}_w[k] &= \bar{a}'\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}[k]^2
\end{aligned}
$$

3 Error increment

$$
\begin{aligned}
y_1[k] - \delta\hat{\bar{w}}_1[k] &= e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] = (1-\lambda_c)\big[e_{\delta\bar{w}_3}[k] - e_{\delta\bar{w}_1}[k]\big] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= \big(\bar{a}_w[k+1] - \bar{a}_w[k]\big) - \big(\hat{\bar{a}}_w[k+1] - \hat{\bar{a}}_w[k]\big) \\
&= \big(\bar{a}' - \bar{a}''\, e_{\delta\bar{w}_3}\big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \tfrac{1}{2}\,\bar{a}''\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2 - \bar{a}'\,\Delta\hat{\bar{w}} - \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}^2 \\
&= \bar{a}'\, e_{\Delta\bar{w}} - \bar{a}''\, e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}} - \bar{a}''\, e_{\delta\bar{w}_3}\, e_{\Delta\bar{w}} + \bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2 + \mathcal{O}(3)
\end{aligned}
$$

4 Mean

$$
\begin{aligned}
\mathbb{E}[e_{\delta\bar{w}_1}] = \mathbb{E}[e_{\delta\bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w] &= 0 \\
\mathbb{E}[e_{\delta\bar{w}_3}^2] = P_{33}, \quad \mathbb{E}[e_{\delta\bar{w}_1}^2] = P_{11}, \quad \mathbb{E}[e_{\delta\bar{w}_1} e_{\delta\bar{w}_3}] &= P_{13} \\
\mathbb{E}[e_{\delta\bar{w}_3}\, e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\delta\bar{w}_1}\, e_{\bar{a}_w}] = P_{14}, \quad \mathbb{E}[e_{\bar{a}_w}^2] &= P_{44} \\
\mathbb{E}[\tilde{\varepsilon}_w\, e_{\delta\bar{w}_1}] = \mathbb{E}[\tilde{\varepsilon}_w\, e_{\delta\bar{w}_3}] = \mathbb{E}[\tilde{\varepsilon}_w\, e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w\, n_w] &= 0 \\
\mathbb{E}\big[e_j\, \delta\hat{\bar{w}}_i\big] = \mathbb{E}\big[e_j\, y_1\big] &= 0 \qquad \text{(posterior error $\perp$ data)}
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}\big[\bar{a}'\, e_{\Delta\bar{w}}\big] &= 0 \\
\mathbb{E}\big[-\bar{a}''\, e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}}\big] &= 0 \\
\mathbb{E}\big[-\bar{a}''\, e_{\delta\bar{w}_3}\, e_{\Delta\bar{w}}\big] &= -\bar{a}''\,\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw}P_{34}\big] \\
\mathbb{E}\big[\bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}}\big] &= 0 \\
\mathbb{E}\big[\tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2\big] &= \tfrac{1}{2}\,\bar{a}''\,\big[(1-\lambda_c)^2\big(P_{33} + P_{11} - 2P_{13}\big) + 2(1-\lambda_c)F_{dw}\big(P_{34} - P_{14}\big) \\
&\qquad + F_{dw}^2 P_{44} + \tilde{Q}_{33}\big]
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = \bar{a}''\Big\{ &-(1-\lambda_c)\big(P_{33} - P_{13}\big) - F_{dw}P_{34} \\
&+ \tfrac{1}{2}(1-\lambda_c)^2\big(P_{33} + P_{11} - 2P_{13}\big) + (1-\lambda_c)F_{dw}\big(P_{34} - P_{14}\big) \\
&+ \tfrac{1}{2}F_{dw}^2 P_{44} + \tfrac{1}{2}\tilde{Q}_{33}\Big\}
\end{aligned}
$$

(first, second and fourth lines: $\bar{a}'$ and $\Delta\hat{\bar{w}}$ are functions of the data, $e_{\Delta\bar{w}}$ and $e_{\delta\bar{w}_3}$ are posterior errors, $\mathbb{E}[\, e \,|\, \text{data}\,] = 0$; without the $y_1 - \delta\hat{\bar{w}}_1$ input $e_{\Delta\bar{w}}$ contains $(1-\lambda_c)n_w$, which is correlated with the data, and these means become the n_w feedthrough shares of 0902_formC_aptrue_4state.tex, Error dynamics, each $\propto \bar{a}''(1-\lambda_c)\ell_{31}R_1$, whose sum was measured to vanish)

5 Estimator with the mean term

$$\begin{aligned}\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \bar{a}'\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}[k]^2 \\ + \bar{a}''\Big\{-(1-\lambda_c)\big(P_{33}[k] - P_{13}[k]\big) - F_{dw}[k]\,P_{34}[k] \\ + \tfrac{1}{2}(1-\lambda_c)^2\big(P_{33}[k] + P_{11}[k] - 2P_{13}[k]\big) \\ + (1-\lambda_c)F_{dw}[k]\big(P_{34}[k] - P_{14}[k]\big) + \tfrac{1}{2}F_{dw}[k]^2 P_{44}[k] + \tfrac{1}{2}\tilde{Q}_{33}[k]\Big\}\end{aligned}$$

$$\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = 0$$

(the braced term is a deterministic input to the predict; it is not differentiated and does not enter F_e, H, Q)

6 Jacobians

$$\frac{\partial\,\delta\hat{\bar{w}}_3[k+1]}{\partial\,\delta\hat{\bar{w}}_3[k]} = \lambda_c, \qquad \frac{\partial\,\delta\hat{\bar{w}}_3[k+1]}{\partial\,\delta\hat{\bar{w}}_1[k]} = (1-\lambda_c)$$

$$\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\delta\hat{\bar{w}}_3[k]} = (1-\lambda_c)\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) - \bar{a}''\,\Delta\hat{\bar{w}}[k] \;\to\; (1-\lambda_c)\,\bar{a}'$$

$$\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\delta\hat{\bar{w}}_1[k]} = -(1-\lambda_c)\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; -(1-\lambda_c)\,\bar{a}'$$

$$\frac{\partial\bar{a}_w[k+1]}{\partial\bar{a}_w[k]} = 1 + F_{dw}[k]\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; 1 + F_{dw}[k]\,\bar{a}'$$

(limits at $\Delta\hat{\bar{w}} \to 0$, the tangent the code keeps; the $-\bar{a}''\Delta\hat{\bar{w}}$ in the first gain-row entry is $\partial\bar{a}'/\partial\delta\hat{\bar{w}}_3 = -\bar{a}''$ of the 'est' reading; the last entry and $\partial\,\delta\bar{w}_3[k+1]/\partial\bar{a}_w[k] = -F_{dw}$ are entries of the TRUE transition, not of the estimator's lines (which do not contain $\hat{\bar{a}}_w$): the truth's $F_{dw}\,e_{\bar{a}_w}$ inside $\Delta\bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{\bar{a}}_w)$)

7 Error dynamics

$$\begin{aligned}e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\ e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] + (1-\lambda_c)\,e_{\delta\bar{w}_1}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \tilde{\varepsilon}_w[k] \\ e_{\bar{a}_w}[k+1] &= -(1-\lambda_c)\,\bar{a}'\,e_{\delta\bar{w}_1}[k] + (1-\lambda_c)\,\bar{a}'\,e_{\delta\bar{w}_3}[k] + \big(1 + F_{dw}\,\bar{a}'\big)\,e_{\bar{a}_w}[k] + \bar{a}'\,\tilde{\varepsilon}_w[k] \\ &\quad - \bar{a}''\,e_{\delta\bar{w}_3}\,\Delta\hat{\bar{w}} - \bar{a}''\,e_{\delta\bar{w}_3}\,e_{\Delta\bar{w}} + \bar{a}''\,\Delta\hat{\bar{w}}\,e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\,e_{\Delta\bar{w}}^2 - \mathbb{E}[\,\cdot\,]\end{aligned}$$

(third line: $\delta\bar{w}_3[k+1] - \delta\hat{\bar{w}}_3[k+1]$ with $y_1 - \delta\hat{\bar{w}}_1 = e_{\delta\bar{w}_1} + n_w$, the n_w of ε_w cancels; fourth line: the second-order group minus its mean of the previous section)

$$\mathbf{e}[k+1] = \tilde{F}_e[k]\,\mathbf{e}[k] + \tilde{\mathbf{q}}[k], \qquad \tilde{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (1-\lambda_c) & 0 & \lambda_c & -F_{dw} \\ -(1-\lambda_c)\,\bar{a}' & 0 & (1-\lambda_c)\,\bar{a}' & 1 + F_{dw}\,\bar{a}' \end{bmatrix}$$

$$\tilde{F}_e[k] = F_e[k] - g_n[k]\,\mathbf{e}_1^\top, \qquad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1-\lambda_c)\,\bar{a}' & 1 + F_{dw}\,\bar{a}' \end{bmatrix}$$

$$g_n[k] = (1-\lambda_c)\begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \qquad \tilde{\mathbf{q}}[k] = \tilde{\varepsilon}_w[k]\begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \qquad \mathbb{E}\big[\tilde{\mathbf{q}}[k]\,n_w[k]\big] = 0$$

$$P^f[k+1] = \tilde{F}_e[k]\,P[k]\,\tilde{F}_e[k]^\top + \tilde{Q}[k]$$

8 Measurement and update

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'\,\nabla_d \bar{w}_d[k] \\ H &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ e_{y_1}[k] &= e^-_{\delta\bar{w}_1}[k] + n_w[k], \qquad e_{y_2}[k] = e^-_{\bar{a}_w}[k] + n_a[k] \end{aligned}$$

$$\begin{aligned} L[k] &= P^f H^\top \big(H P^f H^\top + R\big)^{-1} \\ \hat{\mathbf{x}}[k] &= \hat{\mathbf{x}}^-[k] + L[k]\,\mathbf{e}_y[k] \\ P[k] &= \big(I - L[k]H\big)P^f[k] \\ y_1[k] - \delta\hat{w}_1[k] &= \frac{R_1}{P^f_{11}[k] + R_1}\,e_{y_1}[k] - \ell_{12}[k]\,\big(e_{y_2}[k] - \ell_{41}[k]\,e_{y_1}[k]\big) = \mathbb{E}\big[n_w[k]\,\big|\,y[1..k]\big] \end{aligned}$$

(last line: the posterior residual the Estimator section feeds into the next step, written for the sequential update y_1 then y_2 as in the code; $e_{y_2} - \ell_{41} e_{y_1}$ is the y_2 innovation after the y_1 update, $\ell_{12} = H_{24} P^{(1)}_{14}/S_2$ equals the joint L_{12}; exact in the linear-Gaussian model, approximate for the EKF)

9 Noise

$$\kappa_T = \frac{4k_B T\, a_o}{R}, \qquad \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}$$

$$\tilde{Q}_{33} = \kappa_T \Big(\hat{\hat{a}}_w[k] + (1-\lambda_c)^2 \big(\hat{\hat{a}}_w[k-1] + \hat{\hat{a}}_w[k-2] \big) \Big) \qquad \big(= Q_{33} - (1-\lambda_c)^2 \sigma_{n_w}^2 \big)$$

$$\tilde{Q}_{34} = \tilde{Q}_{43} = -\bar{a}'\, \tilde{Q}_{33}$$

$$\tilde{Q}_{44} = \bar{a}'^2\, \tilde{Q}_{33}$$

$$R = \mathrm{diag}(R_1, R_2), \qquad R_1 = \sigma_{n_w}^2, \qquad R_2 = 2a_{\mathrm{cov}}^2\, IF_{\mathrm{var}}\big(\hat{\hat{a}}_w\big)\big(\hat{\hat{a}}_w + \xi\big)^2$$

(IF_{var}, ξ as in 0902_formC_aptrue_4state.tex; the n_w share $(1-\lambda_c)^2\sigma_{n_w}^2$ has left Q and entered the Estimator section as the input $g_n\,(y_1 - \delta\hat{\hat{w}}_1)$)

10 b_{true} arm: the law read at the estimated gain

(this arm: $b := b_{\text{true}}(\bar w[k-1])$ read at the previous step's TRUE height (code, b_true_at = 'true'), treated as exogenous; slot 5 locked; $\bar a' := b\,\big(1-\hat{\bar a}_w[k]\big)^2$ and $\bar a'' := -2b\big(1-\hat{\bar a}_w[k]\big)\,\bar a'$ read at the ESTIMATED GAIN, not at a height; $\dfrac{\partial \bar a'}{\partial \hat{\bar a}_w} = -2b\big(1-\hat{\bar a}_w\big) = \dfrac{\bar a''}{\bar a'}$; everything not listed is as in Sections 1–9)

True dynamics, the slope line

$$\bar a'_{\text{true}}\big(\bar w[k]\big) = b\,\big(1-\bar a_w[k]\big)^2 = b\,\big(1-\hat{\bar a}_w[k]-e_{\bar a_w}[k]\big)^2 = \bar a' + \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}[k] + \mathcal O\big(e^2\big)$$

$$\bar a_w[k+1]-\bar a_w[k] = \Big(\bar a' + \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}[k]\Big)\,\Delta\bar w[k] + \tfrac12\,\bar a''\,\Delta\bar w[k]^2$$

Error increment (the $-\bar a''\,e_{\delta\bar w_3}$ lines of Section 3 are replaced)

$$e_{\bar a_w}[k+1]-e_{\bar a_w}[k] = \bar a'\,e_{\Delta\bar w} + \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}\,\Delta\hat{\bar w} + \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}\,e_{\Delta\bar w} + \bar a''\,\Delta\hat{\bar w}\,e_{\Delta\bar w} + \tfrac12\,\bar a''\,e_{\Delta\bar w}^2 + \mathcal O(3)$$

Mean

$$\begin{aligned}
\mathbb E\Big[\frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}\,\Delta\hat{\bar w}\Big] &= 0\\
\mathbb E\Big[\frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,e_{\bar a_w}\,e_{\Delta\bar w}\Big] &= \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big]\\
\mathbb E\big[e_{\bar a_w}[k+1]\big]-\mathbb E\big[e_{\bar a_w}[k]\big] &= \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big]\\
&\quad+\tfrac12\,\bar a''\,\big[(1-\lambda_c)^2\big(P_{33}+P_{11}-2P_{13}\big)+2(1-\lambda_c)F_{dw}\big(P_{34}-P_{14}\big)\\
&\qquad+F_{dw}^2P_{44}+\tilde Q_{33}\big]
\end{aligned}$$

Estimator with the mean term

$$\begin{aligned}
\hat{\bar a}_w[k+1] &= \hat{\bar a}_w[k] + \bar a'\,\Delta\hat{\bar w}[k] + \tfrac12\,\bar a''\,\Delta\hat{\bar w}[k]^2\\
&\quad+\frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\big[(1-\lambda_c)\big(P_{34}[k]-P_{14}[k]\big)+F_{dw}[k]\,P_{44}[k]\big]\\
&\quad+\tfrac12\,\bar a''\,\big[(1-\lambda_c)^2\big(P_{33}[k]+P_{11}[k]-2P_{13}[k]\big)+2(1-\lambda_c)F_{dw}[k]\big(P_{34}[k]-P_{14}[k]\big)\\
&\qquad+F_{dw}[k]^2P_{44}[k]+\tilde Q_{33}[k]\big]
\end{aligned}$$

$$\mathbb E\big[e_{\bar a_w}[k+1]\big]-\mathbb E\big[e_{\bar a_w}[k]\big] = 0$$

Difference to the Section 5 term (gain-read minus height-read start point)

$$\begin{aligned}
&\frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\big[(1-\lambda_c)\big(P_{34}-P_{14}\big)+F_{dw}P_{44}\big] + \bar a''\,\big[(1-\lambda_c)\big(P_{33}-P_{13}\big)+F_{dw}P_{34}\big]\\
&\quad= \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\Big[(1-\lambda_c)\big(P_{34}+\bar a'P_{33}-P_{14}-\bar a'P_{13}\big)+F_{dw}\big(P_{44}+\bar a'P_{34}\big)\Big]\\
&\quad= \frac{\partial \bar a'}{\partial \hat{\bar a}_w}\,\mathbb E\Big[\big(e_{\bar a_w}+\bar a'\,e_{\delta\bar w_3}\big)\,e_{\Delta\bar w}\Big]
\end{aligned}$$

(= 0 when the gain error is slaved to the position error through the law, $e_{\bar a_w} = -\bar a'\,e_{\delta\bar w_3}$, i.e. $P_{34} = -\bar a'P_{33}$, $P_{14} = -\bar a'P_{13}$ and $P_{44} = \bar a'^2P_{33}$ (the first two alone leave $F_{dw}(P_{44}-\bar a'^2P_{33})$); hold, both trajectories: $P_{34} + \bar{a}' P_{33} \approx 3 \times 10^{-9}$, the difference is $< 0.005 \times 10^{-6}$ per step; near-wall half of the descent: -3.5×10^{-6} (canon) and -0.5×10^{-6} (Meng) per step in $\mathbb{E}[\hat{\bar{a}}_w - \bar{a}_w]$, running sums -0.0026 and -0.0040 at the start of the hold (code: ctrl_const.pred_mean2_e4, the difference of the two pred_mean2 logs, 10 seeds, run_btrue_e4.m); the closed-loop response to this input is smaller than its running sum, $+0.0010$ (canon) and $+0.0004$ (Meng) at the end of the descent, relaxing to < 0.0003 within the following second)

(treating b as exogenous drops the mean $-(1-\hat{\bar{a}}_w)^2\, b'_{\mathrm{true}}\, \mathbb{E}\big[e_{\delta\bar{w}_3}[k-1]\, e_{\Delta\bar{w}}\big]$ of $\bar{a}'\, e_{\Delta\bar{w}}$, b being read at the true height $\bar{w}_d[k-1] - \delta\bar{w}_3[k-1]$; relative to the kept $\bar{a}''$ term it is $\sim b'_{\mathrm{true}}/\big(2b^2(1-\hat{\bar{a}}_w)\big)$, a few percent, b_{true} spanning 8/9 to 1 over the band)

Jacobians and error dynamics (the entries that change)

$$
\begin{aligned}
\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \delta\hat{\bar{w}}_3[k]} &= (1-\lambda_c)\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; (1-\lambda_c)\,\bar{a}' \\
\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \hat{\bar{a}}_w[k]} &= 1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\Delta\hat{\bar{w}}[k] + F_{dw}[k]\big(\bar{a}' + \bar{a}''\,\Delta\hat{\bar{w}}[k]\big) \;\to\; 1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\Delta\hat{\bar{w}}[k] + F_{dw}[k]\,\bar{a}' \\
\tilde{F}_e(4,4) &= 1 + F_{dw}\,\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\Delta\hat{\bar{w}} \\
y_2[k] &= \bar{a}_w[k] - \bar{a}'\,\nabla_d \bar{w}_d[k] + n_a[k], \qquad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\,\nabla_d \bar{w}_d[k]
\end{aligned}
$$

(no $-\bar{a}''\Delta\hat{\bar{w}}$ in the $\delta\hat{\bar{w}}_3$ entry: the slope does not read a height; $H_{24} \neq 1$ whenever $\bar{a}'$ reads $\hat{\bar{a}}_w$ (Sections 10, 11; code line `1 - Grad_wbar_d*A_a_i`), $= 1$ only for the exogenous slope of Sections 1–9; the $\partial\bar{a}'/\partial\hat{\bar{a}}_w\,\Delta\hat{\bar{w}}$ entry is the law's own amplification of a gain error along the step, > 1 when moving toward the wall; $\tilde{Q}_{34} = -\bar{a}'\tilde{Q}_{33}$, $\tilde{Q}_{44} = \bar{a}'^2\tilde{Q}_{33}$ unchanged)

11 Production: b a constant state

(this arm: $\hat{b} := \hat{x}_5$, constant model, $\tilde{Q}_{55} = 0$, seed 8/9, $\sqrt{P_{55}[0]} = \sup_{\text{env}} |b_{\text{true}} - 8/9|$; $\bar{a}' := \hat{b}\,(1-\hat{\bar{a}}_w)^2$, $\dfrac{\partial \bar{a}'}{\partial \hat{b}} = (1-\hat{\bar{a}}_w)^2 = \dfrac{\bar{a}'}{\hat{b}}$, $\bar{a}'' := -2\hat{b}(1-\hat{\bar{a}}_w)\,\bar{a}'$; $e_b[k] := b_{\text{true}}\big(\bar{w}[k]\big) - \hat{b}[k]$, NOT zero-mean: b_{true} is a curve, $\hat{b}$ a constant; $f(\bar{w}) := \big(1-\bar{a}_{\text{true}}(\bar{w})\big)^2\big(b_{\text{true}}(\bar{w}) - \hat{b}\big)$, $F' = f$; everything not listed is as in Sections 1–10)

True dynamics, slope and curvature

$$
\begin{aligned}
\bar{a}'_{\text{true}}\big(\bar{w}[k]\big) &= b_{\text{true}}\big(\bar{w}[k]\big)\big(1-\bar{a}_w[k]\big)^2 = \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, e_{\bar{a}_w}[k] + \frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b[k] + \mathcal{O}\big(e^2\big) \\
\bar{a}''_{\text{true}}\big(\bar{w}[k]\big) &= \frac{d}{d\bar{w}}\Big[b_{\text{true}}(\bar{w})\big(1-\bar{a}_{\text{true}}(\bar{w})\big)^2\Big] = \bar{a}'' + f'\big(\bar{w}[k]\big) + \frac{\bar{a}''}{\bar{a}'}\, f\big(\bar{w}[k]\big) + \mathcal{O}(e_{\bar{a}_w})
\end{aligned}
$$

($\bar{a}''/\bar{a}' = -2\hat{b}(1-\hat{\bar{a}}_w)$; the $(\bar{a}''/\bar{a}')f$ piece is first order in e_b, the same order as the model-error line, and is kept)

$\hat{b}$ and e_b

$$
\begin{aligned}
\hat{b}^-[k+1] &= \hat{b}[k] \\
\hat{b}[k+1] &= \hat{b}^-[k+1] + \ell_{51}[k+1]\, e_{y_1}[k+1] + \ell_{52}[k+1]\, e_{y_2}[k+1] \\
e_b[k+1] &= e_b[k] + b'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] - \ell_{51}[k+1]\, e_{y_1}[k+1] - \ell_{52}[k+1]\, e_{y_2}[k+1]
\end{aligned}
$$

Error increment (Section 10 plus the e_b and f' lines)

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] = \bar{a}'\, e_{\Delta\bar{w}} &+ \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, e_{\bar{a}_w}\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big) + \bar{a}''\,\Delta\hat{\bar{w}}\, e_{\Delta\bar{w}} + \tfrac{1}{2}\,\bar{a}''\, e_{\Delta\bar{w}}^2 \\
&+ \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}\, f\Big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2 + \mathcal{O}(3)
\end{aligned}
$$

Mean, the e_b and curvature lines (P_{i5} := the filter's own posterior cross-covariance under its constant-b model; the part of e_b that fluctuates given the data is $-b'_{\text{true}}\, e_{\delta\bar{w}_3}$, which is NOT P_{i5})

$$
\begin{aligned}
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\,\Delta\hat{\bar{w}}\Big] &= \frac{\partial \bar{a}'}{\partial \hat{b}}\,\mathbb{E}[e_b]\,\Delta\hat{\bar{w}} \qquad \text{(first order; } \mathbb{E}[e_b] = b_{\text{true}}(\bar{w}) - \mathbb{E}[\hat{b}] \neq 0\text{)} \\
\mathbb{E}\Big[\frac{\partial \bar{a}'}{\partial \hat{b}}\, e_b\, e_{\Delta\bar{w}}\Big] &= \frac{\partial \bar{a}'}{\partial \hat{b}}\,\big[(1-\lambda_c)\big(P_{35} - P_{15}\big) + F_{dw}P_{45}\big] \\
&\quad - \frac{\partial \bar{a}'}{\partial \hat{b}}\, b'_{\text{true}}\,\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw}P_{34}\big] \\
\mathbb{E}\Big[\tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}f\Big)\big(\Delta\hat{\bar{w}} + e_{\Delta\bar{w}}\big)^2\Big] &= \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}f\Big)\big(\Delta\hat{\bar{w}}^2 + \mathbb{E}[e_{\Delta\bar{w}}^2]\big)
\end{aligned}
$$

(second line: the first bracket is the filter's cross-covariance with its constant unknown b, the second is the height dependence of b_{true} through $e_{\delta\bar{w}_3}$; b'_{true} is not c-free)

$$
\begin{aligned}
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = &\ \frac{\partial \bar{a}'}{\partial \hat{b}}\,\mathbb{E}[e_b]\,\Delta\hat{\bar{w}} + \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'}f\Big)\big(\Delta\hat{\bar{w}}^2 + \mathbb{E}[e_{\Delta\bar{w}}^2]\big) \\
&- \frac{\partial \bar{a}'}{\partial \hat{b}}\, b'_{\text{true}}\big[(1-\lambda_c)\big(P_{33} - P_{13}\big) + F_{dw}P_{34}\big] \\
&+ \frac{\partial \bar{a}'}{\partial \hat{b}}\,\big[(1-\lambda_c)\big(P_{35} - P_{15}\big) + F_{dw}P_{45}\big] + \text{(Section 10 lines)}
\end{aligned}
$$

(first two lines: model error, needs b_{true} and b'_{true}, not c-free, not added; third line: added, the Section 10 term with e_b)

Estimator with the mean term (the line added to Section 10's)

$$\hat{a}_w[k+1] = (\text{Section 10}) + \frac{\partial \bar{a}'}{\partial \hat{b}} \left[(1-\lambda_c)\big(P_{35}[k] - P_{15}[k]\big) + F_{dw}[k]\, P_{45}[k]\right]$$

Model error along a path (first order in e_b; open loop, $\mu := \mathbb{E}[e_{\bar{a}_w}]_{\text{model}}$)

$$\mu[k+1] = \Big(1 + \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w} \Delta\bar{w}[k]\Big)\mu[k] + f\big(\bar{w}[k]\big)\, \Delta\bar{w}[k] + \tfrac{1}{2}\Big(f' + \frac{\bar{a}''}{\bar{a}'} f\Big)\Delta\bar{w}[k]^2 + \mathcal{O}(3)$$

$$\frac{d\mu}{d\bar{w}} = \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w}\, \mu + f \qquad \Longrightarrow \qquad \mu(\bar{w}) = \big(1 - \bar{a}_{\text{true}}(\bar{w})\big)^2 \int_{\bar{w}_{\text{start}}}^{\bar{w}} \big(b_{\text{true}} - \hat{b}\big)\, d\bar{w}'$$

$$\frac{1}{1-\bar{a}_{\text{true}}(\bar{w})} - \frac{1}{1-\hat{\bar{a}}_w(\bar{w})} = \int_{\bar{w}_{\text{start}}}^{\bar{w}} \big(b_{\text{true}} - \hat{b}\big)\, d\bar{w}' \qquad \text{(exact, noise-free: } \tfrac{d}{d\bar{w}} \tfrac{1}{1-\bar{a}} = b\text{)}$$

$$\text{stationary hold:} \quad \mathbb{E}\big[\Delta\mu\big] \to 0$$

$$\mathbb{E}[f\, \Delta\bar{w}] = -(1-\lambda_c)\, f'\, \mathrm{Var}(\bar{w}), \qquad \tfrac{1}{2} f'\, \mathbb{E}[\Delta\bar{w}^2] = +(1-\lambda_c)\, f'\, \mathrm{Var}(\bar{w})$$

$$\tfrac{1}{2}\tfrac{\bar{a}''}{\bar{a}'} f\, \mathbb{E}[\Delta\bar{w}^2] + \tfrac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w} \mathbb{E}[\Delta\bar{w}\, e_{\bar{a}_w}] = 0 \qquad \text{(likewise)}$$

(the forcing f alone integrates to $F(\bar{w}_{\text{end}}) - F(\bar{w}_{\text{start}})$, $F' = f$; the law then AMPLIFIES what was injected far away by $[(1-\bar{a}_{\text{wall}})/(1-\bar{a}_{\text{far}})]^2 = 29.8$ from $\bar{w} = 6.67$ to 1.10 – the open loop of the measured $\times 22$)

($\hat{b} = 8/9$, $c_\perp$ of the plant: $b_{\text{true}}(6.67) = 0.883$, minimum 0.867 at $\bar{w} = 2.19$, crosses 8/9 at $\bar{w} = 1.325$, $b_{\text{true}}(1.10) = 0.928$; $f(1.10) = +0.032$, $f(1.8) = -0.0066$, $f(6.67) = -0.0002$. Open loop, $\bar{a}_w - \hat{\bar{a}}_w$ at the wall: Meng starts at $\bar{w} = 6.67$ (15 μm): $\int_{6.67}^{1.10} (b_{\text{true}} - 8/9)\, d\bar{w} = +0.068$, $\mu(1.10) = (1 - 0.0873)^2 \times 0.068 = +0.057$ (exact law integration +0.060, 69% of $\bar{a}_w(1.10)$); canon starts at $\bar{w} = 22.2$ (50 μm): +0.084 (+0.093 exact). The estimate arrives BELOW the truth, the long stretch with $b_{\text{true}} < 8/9$ outweighing the short one above it; the 09-04 first writing integrated f alone and gave 0.008 – FAIL caught by the independent verification. MEASURED (run_prod_ladder.m, four blocks, 10 seeds, paired to the b_{true}-curve arm): b locked at 8/9: end of descent -0.0030 ± 0.0002 (canon), -0.0005 ± 0.0002 (Meng); hold start -0.0005, -0.0009; the y_2 leg in motion removes $\geq 95\%$ of the open-loop error, sign as derived. b locked at $b_{\text{true}}(\bar{w}_{\text{hold}}) = 0.928$ (oracle): end of descent -0.0066, -0.0017 (more negative: $\hat{b} > b_{\text{true}}$ over the whole mid range, as $f < 0$ says). $\hat{b}$ estimated from 8/9 (production): minus locked 8/9 at the hold start $+0.0014 \pm 0.0009$ (canon), -0.00002 ± 0.00013 (Meng), $\hat{b}$ stays within 0.01 of the seed (0.885 ± 0.017, 0.896 ± 0.021 at the end), $\sqrt{P_{55}}$ 0.039 $\to$ 0.023, 0.027. Absolute hold level of production with the four blocks: $+0.0012 \pm 0.0016$ (canon), $+0.0004 \pm 0.0023$ (Meng); production as it was (Euler, no mean terms): +0.019, +0.010. OPEN: a paired hold-slope difference constant-b minus curve of -0.176 ± 0.025 (canon), -0.184 ± 0.015 (Meng) $\times 10^{-6}$ per step; not the b'_{true} cross-moment ($< 0.002 \times 10^{-6}$ on the filter's P, probe_lockb_bprime_line.m); with b locked at the hold value it vanishes on canon (-0.006 ± 0.032) but not on Meng (-0.165 ± 0.014), so it is not simply the slope mismatch at the hold height either; -0.0014 over a 5 s hold, mechanism open)

Jacobians and measurement (the entries that change; code: state writing, $\partial \bar{a}'/\partial \hat{b} = +\bar{a}'/\hat{b}$)

$$\frac{\partial \hat{\bar{a}}_w[k+1]}{\partial \hat{b}[k]} = \frac{\partial \bar{a}'}{\partial \hat{b}}\, \Delta\hat{\bar{w}}[k], \qquad \frac{\partial \hat{b}[k+1]}{\partial \hat{b}[k]} = 1$$

$$y_2[k] = \bar{a}_w[k] - \bar{a}'\, \nabla_d \bar{w}_d[k] + n_a[k], \qquad H_{24} = 1 - \frac{\partial \bar{a}'}{\partial \hat{\bar{a}}_w} \nabla_d \bar{w}_d[k], \qquad H_{25} = -\frac{\partial \bar{a}'}{\partial \hat{b}} \nabla_d \bar{w}_d[k]$$

$$g_n[k] = (1-\lambda_c)\, [\,0,\ 0,\ -1,\ \bar{a}',\ 0\,]^\top, \qquad \tilde{Q}_{5j} = \tilde{Q}_{j5} = 0$$

(both $\hat{b}$ columns carry a displacement factor and vanish together in a hold, Section 3 of observability-workflow; locked-b arm: the filter's $P_{i5} = 0$, so the added line vanishes and the model-error lines, including the b'_{true} cross-moment, remain)

The container of the $b'_{\text{true}}\,\Delta\bar{w}$ line: $\tilde{Q}_{55}$ proportional to the path (09-07)

(e_b is driven every step by $b'_{\text{true}}(\bar{w})\,\Delta\bar{w}$, the change of the exact law factor along the path; the constant model has no term for it and $\tilde{Q}_{55} = 0$ lets P_{55} collapse below $|e_b|$ – measured 09-07 on the estimated-$\hat{b}$ arm with the seed-at-truth $P_{44}[0]$: $\sqrt{P_{55}} \to 0.0065$, hold -0.0106 ± 0.0021 canon, $\hat{b}$ pinned at 0.96 against $b_{\text{true}}(\bar{w}_{\text{hold}}) = 0.928$. $\Delta b := \sup_{\text{env}} b_{\text{true}} - \inf_{\text{env}} b_{\text{true}}$, $W := \bar{w}_{hi} - \bar{w}_{lo}$, both from the envelope the driver already sweeps; the sign of b'_{true} is not known to the filter)

$$e_b[k+1] = e_b[k] + b'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] - \ell_{51}[k+1]\,e_{y_1}[k+1] - \ell_{52}[k+1]\,e_{y_2}$$

$$\sum_{\text{one traverse}} \big|\, b'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k]\,\big| \le \Delta b$$

$$\tilde{Q}_{55}[k] = \frac{\Delta b^2}{W}\,\big|\Delta\hat{\bar{w}}[k]\big|$$

$$\sum_{\text{one traverse}} \tilde{Q}_{55}[k] = \Delta b^2$$

$$P^-_{55}[k+1] = P_{55}[k] + \tilde{Q}_{55}[k] \qquad \text{(row and column 5 of } \tilde{F}_e P \tilde{F}_e^\top \text{ unchanged)}$$

$$\sqrt{P_{55}} \text{ after a path } sW,\ 0 \le s \le 1,\ \text{from } P_{55} = 0 = \Delta b\,\sqrt{s}\ \ge\ \Delta b\, s\ \ge\ \Big|\sum b'_{\text{true}}\Delta\bar{w}\Big| \qquad \text{only if } \max\big|b'_{\text{true}}\big| \le \Delta b/$$

$$\tilde{Q}_{55} = 0 \text{ in a hold } (\Delta\hat{\bar{w}} = 0), \qquad \Delta b = 0 \text{ on the } \mathrm{b}_{\text{true}} \text{ arm (Section 1}$$

(no free number: Δb and W are the envelope's; $\Delta b^2/W$ is passed by the driver as q55_per_R, the controller multiplies by $|\Delta\hat{\bar{w}}|$ each step. Code: ctrl_const.q55_path (default false). Acceptance pre-registered in test_script/scratch/run_ladder_endpoints.m arms bhq5 / btq5 / bhq5s: the estimated-$\hat{b}$ arm with the seed-at-truth $P_{44}[0]$ must reach hold $|\text{mean}| \le 1$ SEM and a near-wall mean $|\cdot| < 0.01$ (from -0.0106 and -0.04), $\sqrt{P_{55}}$ at the end ≥ 0.02, $\hat{b}$ in hold within $\sqrt{P_{55}}$ of $b_{\text{true}}(\bar{w}_{\text{hold}})$, near-wall $\sigma_{\text{seed}} \le 0.015$; the b_{true} arm bit-identical; production ($P_{44}[0]$ as is) with the term: hold within 1 SEM, σ_{seed} not above 0.032)

Measured (10 seeds, four blocks, $\kappa = 1$; Meng [canon]; near-wall window Meng 7–10 s, canon 1.0–1.5 s; $\bar{a}$ units)

seed-at-truth $P_{44}[0]$, $\tilde{Q}_{55} = 0$:	hold -0.0024 ± 0.0021 [-0.0106 ± 0.0021],	near wall $-0.019\ \sigma\, 0.017$ [-0.012
seed-at-truth $P_{44}[0]$, $\tilde{Q}_{55}$ on :	hold -0.0006 ± 0.0024 [$+0.0059 \pm 0.0031$],	near wall $-0.013\ \sigma\, 0.039$ [-0.003
production $P_{44}[0]$, $\tilde{Q}_{55}$ on :	hold -0.0006 ± 0.0024 [$+0.0060 \pm 0.0031$],	near wall $-0.012\ \sigma\, 0.039$ [-0.003
b_{true} arm with the term :	bit-identical to Section 10,	$b_{\text{true}}(\bar{w}_{\text{hold}}) = 0.924$ [0.928]

(STATUS 09-07: REFUTED as written. The container removes the hold bias on Meng and halves canon's near-wall mean, but doubles the near-wall spread (pre-registered lines 2 and 4 fail), $\sqrt{P_{55}}$ over-covers $|e_b|$ by 2× – the $\sqrt{s}$ against s of the line above, by construction – and $\hat{b}$ drifts below the curve (0.87 against 0.92–0.93) while the extra freedom is filled by gain transients, not by b_{true} (the 08-17 information bound). With the term on, the seed-at-truth and the production $P_{44}[0]$ give the same run: the container dominates P_{44} through $\tilde{F}_e(4,5)$. What the three estimated-$\hat{b}$ rows share is a near-wall MEAN of -0.01 to -0.02 that no covariance container removes – it is the mean model error of a constant b (the μ of this section), and the b_{true} arm, whose mean model follows the curve, has none. A covariance term is the wrong category for a coherent bias; the mean model is. Flag kept, default false. Independent verification 09-07 (sympy / numeric, three interpolants of the real b_{true} through 0.883 at 6.67, 0.867 at 2.19, 0.928 at 1.10): items 1 and 4 exact; the honesty line needs the uniform-rate condition written above – with it dropped a curve that spends its whole Δb early is uncovered; on the real curve the container covers every DESCENT (worst $|e_b|/\sqrt{P_{55}} = 0.74$, at the wall end) but not an ASCENT (2.5× Meng, 4.8× canon, 1.55× the outward oscillation leg): the steep part of b_{true} sits at the wall, late on a descent and early on an ascent. A run that never resets P_{55} carries the descent's margin through the ascent with $\hat{b}$ frozen, but the filter shrinks P_{55} at every update, so that margin is not guaranteed – consistent with the measured $\hat{b}$ drift on the oscillating canon profile.)

E2 (09-07): the additive gain disturbance δa alongside $\hat{b}$ – paper gates 1–3

(slot 6 of the state, the formC_dist 'gain' placement: $\hat{\bar{a}}_w[k+1]$ gains $+\delta\hat{a}[k]$ outside the law bracket, $\delta\hat{a}[k+1] = \delta\hat{a}[k]$, $\sqrt{P_{66}[0]}$ = the driver's $S3(b)$ closed-form supremum, $\tilde{Q}_{66} = 0$, seed 0; purpose: the innovations that the estimated-$\hat{b}$ arm with a small $P_{44}[0]$ charges to $\hat{b}$ – canon +0.057 in the descent, 99% through ℓ_{51}; Meng +0.039 through $P_{54}H_{24}$ – get a level-type state to go to; motivation and the E1 discriminator in the memory file of 09-07)

Gate 1, the two columns of each slot

$$\tilde{F}_e(4,6) = 1, \qquad H_{26} = -d \qquad \text{(structural constants: non-zero in a hold and at a turning point)}$$
$$\tilde{F}_e(4,5) = \frac{\bar{a}'\,\Delta\hat{\bar{w}}}{\hat{b}}, \qquad H_{25} = -\frac{\partial \bar{a}'}{\partial \hat{b}}\,\nabla_d \bar{w}_d \qquad \text{(both carry a displacement)}$$

Gate 2, time signature of an error: $e_{\delta a}$ = a ramp of $\hat{\bar{a}}_w$ with slope $e_{\delta a}$ per step plus the direct readout $-d\,e_{\delta a}$; e_b = a ramp with slope $\bar{a}'\Delta\hat{\bar{w}}\,e_b/\hat{b}$ per step plus the readout $H_{25}e_b$: the same shape wherever $\bar{a}'\Delta\hat{\bar{w}}$ is constant.

Gate 3, the three-state subsystem $[\bar{a}_w,\ \delta a,\ b]$ seen by y_2 ($\nabla_d \bar{w}_d = 0$ for the moment)

$$F_3 = \begin{bmatrix} 1 & 1 & \bar{a}'\Delta\hat{\bar{w}} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad H_3 = \begin{bmatrix} 1 & -d & 0 \end{bmatrix}$$
$$O_3 = \begin{bmatrix} H_3 \\ H_3F_3[k] \\ H_3F_3[k+1]F_3[k] \end{bmatrix} = \begin{bmatrix} 1 & -d & 0 \\ 1 & 1-d & \bar{a}'\Delta\hat{\bar{w}}[k] \\ 1 & 2-d & \bar{a}'\Delta\hat{\bar{w}}[k] + \bar{a}'\Delta\hat{\bar{w}}[k+1] \end{bmatrix}$$
$$\det O_3 = \bar{a}'\Delta\hat{\bar{w}}[k+1] - \bar{a}'\Delta\hat{\bar{w}}[k]$$
$$\det O_3 = 0 \iff \bar{a}'\Delta\hat{\bar{w}} \text{ constant across the window} \qquad \text{(uniform far-field descent: } \delta a \text{ and } b \text{ alias)}$$

(they separate where $\bar{a}'$ changes – near the wall – or where $\Delta\hat{\bar{w}}$ changes – turning points and the entry to a hold; in a hold δa stays observable through $H_{26} = -d$ and the ramp, b does not (Section 11). Gates 4–5: test_script/scratch/verify_obs_da_slot.m, negative control slot 7. Code: ctrl_const.da_slot, default false; arms bhd / btd / bhatd in run_ladder_endpoints.m)

Gates 4–5 measured (seed 7, window 500 steps unless stated; best CRLB / prior; test_script/scratch/verify_obs_d

$$\begin{aligned} \text{canon, } \delta a \text{ free}: &\quad \text{rank } 8/8,\ \text{negative control } \infty\ 15/15; \quad \bar{a}_w\ 0.060,\ b\ 0.047,\ \delta a\ 0.0044 \quad (\text{without } \delta a: \ \bar{a}_w\ 0 \\ \text{Meng, } \delta a \text{ free}: &\quad \text{rank } 8/8,\ \infty\ 40/40; \quad b\ 2.88 \text{ FAIL at 500 steps},\ 0.17 \text{ at } 2000,\ 0.040 \text{ at } 4000; \quad \delta a\ 0.53 \\ \text{gate-3 condition seen}: &\quad \text{on the slow ramp a 500-step window spans } 0.13\,R,\ \bar{a}'\Delta\hat{\bar{w}} \text{ hardly changes} \Rightarrow \delta a \text{ and } b \text{ alia} \end{aligned}$$

Arms (10 seeds; near-wall window Meng 7–10 s, canon 1.0–1.5 s; $\bar{a}$ units; Meng [canon])

bhp0 (estimated $\hat{b}$, seed-at-truth $P_{44}[0]$) :	hold $-\,0.0024 \pm 0.0021$ [-0.0106 ± 0.0021], near wall $-\,0.019$ $\sigma\,0.01$
bhd (the same $+\,\delta a$) :	hold $-\,0.0006 \pm 0.0023$ [$+0.0035 \pm 0.0017$], near wall $-\,0.010$ σ **0.04**
btd ($\mathrm{b_{true}} + \delta a$) :	hold $+\,0.0002 \pm 0.0023$ [$+0.0017 \pm 0.0017$], σ 0.046 [0.038] against b
bhatd (production $P_{44}[0] + \delta a$) :	identical to bhd on both trajectories

(STATUS 09-07: the slot does what it was asked – $\hat{b}$ is no longer pulled to the wall value and locked (0.87–0.89 with $\sqrt{P_{55}}$ open), the canon hold bias -0.0106 becomes $+0.0035$, Meng's -0.0006 – but the pre-registered spread line fails by 3–4×: δa is a per-step INCREMENT with the $S3(b)$ prior 3×10^{-4} [1.6×10^{-5}] per step, which integrates to a gain uncertainty of order 0.1–0.5 over the far-field part of the descent; the filter lets $\hat{\bar{a}}_w$ walk until y_2 pins δa (figure: seed 7 steps of ± 0.05 between 2 and 7 s on Meng, 0.2–1.3 s on canon), and the $\mathrm{b_{true}}$ arm pays the same 0.04. The near-wall MEAN of the estimated-$\hat{b}$ arms is unchanged on Meng (-0.010): as with $\tilde{Q}_{55}$, a state that carries model error does not remove the mean of a constant-b law. REFUTED as a production candidate; kept as an instrument, default false. What survives from E1/E2: the innovations of the estimated-$\hat{b}$ arm with a small $P_{44}[0]$ need a home – a drift state gives them one at the price of far-field spread, a large $P_{44}[0]$ (production) gives them one at the price of near-wall spread, and neither touches the mid-band mean.)

12 Covariance propagation of the self-referential law: the weight κ

(arms whose slope reads $\hat{\bar{a}}_w$, Sections 10–11; $A := \partial\bar{a}'/\partial\hat{\bar{a}}_w = -2\hat{b}(1-\hat{\bar{a}}_w)$; $z := e_{\bar{a}_w} + \bar{a}'\, e_{\delta\bar{w}_3}$ the law-slaved combination of Section 2 of the update-half document; everything else as in Sections 1–11)

Linear error dynamics in the $(e_{\delta\bar{w}_3},\ z)$ basis (from Section 10, $e_{\bar{a}_w} = z - \bar{a}' e_{\delta\bar{w}_3}$, $\bar{a}'[k+1] = \bar{a}' + \bar{a}''\Delta\hat{\bar{w}}$, $\bar{a}'' = A\,\bar{a}'$)

$$
\begin{aligned}
z[k+1] &= \big(1 + A\,\Delta\hat{\bar{w}}\big)\, z[k] - A\,\Delta\hat{\bar{w}}\,\bar{a}'\, e_{\Delta\bar{w}}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \big(\lambda_c + F_{dw}\,\bar{a}'\big)\, e_{\delta\bar{w}_3}[k] + (1-\lambda_c)\, e_{\delta\bar{w}_1}[k] - F_{dw}\, z[k] - \tilde{\varepsilon}_w[k] \\
\ell_{41} + \bar{a}'\,\ell_{31} = 0 &\iff \mathrm{Cov}\big(z,\ e_{\delta\bar{w}_1}\big) = 0
\end{aligned}
$$

(the amplification $1 + A\Delta\hat{\bar{w}}$ acts on z; the position noise enters the gain through the law as $\bar{a}'\,\tilde{\varepsilon}_w$ and is removed by the y_1 update exactly when the second line holds – the $\mathrm{a}'_{\mathrm{true}}$ arm keeps it in every segment, $|\ell_{41} + \bar{a}'\ell_{31}| < 0.002$)

Measured ($\mathrm{b}_{\mathrm{true}}$ arm, four blocks, canon 1.0–1.5 s, 10–30 seeds; Meng 7–10 s in brackets)

$$
\begin{aligned}
\kappa = 1: \quad & P_{44} \text{ grows } \times 16 \text{ along the descent} \\
& \ell_{41} = +0.036, \qquad -\bar{a}'\ell_{31} = -0.157, \qquad \ell_{41} + \bar{a}'\ell_{31} = +0.19\ [+0.039] \\
& \sigma(\text{predict}) = 0.025, \qquad \sigma(y_1 \text{ leg}) = 0.009, \qquad \text{corr} = -0.40 \qquad (\mathrm{a}'_{\mathrm{true}} \text{ arm: } 0.018,\ 0.017,\ -0.99) \\
& \sigma_{\mathrm{seed}}(\hat{\bar{a}}_w - \bar{a}_w) = 0.026\ [0.032] \text{ at the worst instant}, \qquad \sigma_{\mathrm{seed}}/\sqrt{P_{44}} = 0.66\ [0.81] \\
\kappa = 0: \quad & \ell_{41} + \bar{a}'\ell_{31} = 0.002\ [0.001], \qquad \sigma_{\mathrm{seed}} = 0.0045\ [0.0079] \\
& \sigma_{\mathrm{seed}}/\sqrt{P_{44}} = 1.24\ [1.60]\ (\text{fast}), \qquad 1.75\ [2.07]\ (\text{hold})
\end{aligned}
$$

Weighted propagation (covariance only; the state predict, the mean term and H_2 keep A)

$$
\tilde{F}_e(4,4) = 1 + F_{dw}\,\bar{a}' + \kappa\, A\,\Delta\hat{\bar{w}}, \qquad \kappa \in [0,1]
$$

Calibration (seeds 1:10, both trajectories; honesty $= \sigma_{\mathrm{seed}}/\sqrt{P_{44}}$, geometric mean over $\{\text{fast},\ \text{hold}\} \times \{\text{canon},\ \text{Meng}\}$)

$$
\begin{aligned}
\kappa &= 0,\ 0.25,\ 0.5,\ 0.75,\ 1 \\
\text{honesty (geo. mean)} &= 1.64,\ 1.27,\ \mathbf{0.96},\ 0.81,\ 0.84 \\
\sigma_{\mathrm{seed}} \text{ worst instant, canon} &= 0.0045,\ 0.0046,\ \mathbf{0.0054},\ 0.0109,\ 0.0264 \\
\sigma_{\mathrm{seed}} \text{ worst instant, Meng} &= 0.0079,\ 0.0101,\ \mathbf{0.0158},\ 0.0249,\ 0.0323 \\
\text{hold honesty canon / Meng} &= 1.75/2.07,\ 1.68/1.25,\ \mathbf{1.19/1.06},\ 0.88/1.07,\ 0.87/1.08
\end{aligned}
$$

Validation (seeds 11:20, $\kappa = 0.5$ vs 1)

$$
\begin{aligned}
\sigma_{\mathrm{seed}} \text{ worst instant}: &\ 0.0237 \to 0.0026 \text{ (canon)}, \quad 0.0272 \to 0.0082 \text{ (Meng)} \\
\text{hold } \sigma_{\mathrm{seed}}: &\ 0.0103 \to 0.0056, \quad 0.0091 \to 0.0081 \qquad \text{hold honesty}: 1.43 \to 1.25, \quad 1.36 \to 1.29 \\
\text{hold level}: &\ +0.0022 \pm 0.0031 \to +0.0012 \pm 0.0016, \quad +0.0001 \pm 0.0027 \to +0.0004 \pm 0.0024
\end{aligned}
$$

(κ = code ctrl_const.fe44_Aa_scale. STATUS: $\kappa = 0.5$ is a CALIBRATED value for the $\mathrm{b}_{\mathrm{true}}$ arm only (observable: the honesty ratio; calibration and judgement on disjoint seeds). It was made the production default for a few hours on 2026-09-04 and REVERTED to $\kappa = 1$ the same night: on the arms with a constant b the smaller P_{44} lets the constant-b model error of Section 11 stand – b locked at 8/9, canon, 10 seeds: hold -0.0197 ± 0.0007, every seed at the $\bar{a}$ floor ($+0.0006$ at $\kappa = 1$); $\hat{b}$ estimated (production): hold $+0.0030$ vs $+0.0017$, worst-instant mean -0.031 vs -0.034 (no gain), $\hat{b}$ pulled to 0.852–0.858 with $\sqrt{P_{55}}$ collapsing to 0.004 (canon). The calibration does not transfer across arms: the term it weights competes with the y_2 correction of the model error, which the $\mathrm{b_{true}}$ arm does not have. What it stands in for: the split of the gain error into a free part, amplified by $1 + A\Delta\hat{\bar{w}}$, and a law-slaved part that follows $e_{\delta\bar{w}_3}$ and the growth of $\bar{a}'$, which the first-order propagation treats alike; the closed form is open, and a production-safe version must be derived (or calibrated) with the model error present. Not part of this: the y_2 weight – $\mathrm{NIS}_2 = e_{y_2}^2/S_2 = 0.27$–$0.34$ in every segment, both trajectories, both κ (the IF colour design), $\mathrm{NIS}_1 = 1.00$; nothing segment-specific to correct there)

Closed form: the Jacobian of the exact law step (replaces κ)

(the predict of Section 5 with the exact step of 09-01/09-04; $\hat{b}$ = the constant the law uses ($b_{\mathrm{true}}(\bar{w})$ fed on the $\mathrm{b_{true}}$ arm, $\hat{b}$ on the production arm); $\Delta\hat{\bar{w}} = \Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3 + \alpha\,(m_1+m_2) + (1-\lambda_c)\,(y_1 - \delta\hat{\bar{w}}_1)$ the bracket the predict multiplies; $\partial\Delta\hat{\bar{w}}/\partial\hat{\bar{a}}_w = F_{dw}$ the loop coupling of Section 6; $A = \partial\bar{a}'/\partial\hat{\bar{a}}_w = -2\hat{b}(1-\hat{\bar{a}}_w)$)

Exact step and its partial derivatives

$$\hat{\bar{a}}_w[k+1] = 1 - \Big(\frac{1}{1-\hat{\bar{a}}_w[k]} + \hat{b}\,\Delta\hat{\bar{w}}\Big)^{-1}$$

$$\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\hat{\bar{a}}_w[k]} = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2} = \frac{1}{\big(1+\hat{b}\,(1-\hat{\bar{a}}_w[k])\,\Delta\hat{\bar{w}}\big)^2} = 1 + A\,\Delta\hat{\bar{w}} + \tfrac{3}{4}\,A^2\Delta\hat{\bar{w}}^2 + \dots$$

$$\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\Delta\hat{\bar{w}}} = \hat{b}\,(1-\hat{\bar{a}}_w[k+1])^2 = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\bar{a}'$$

$$\frac{\partial\hat{\bar{a}}_w[k+1]}{\partial\hat{b}} = \Delta\hat{\bar{w}}\,(1-\hat{\bar{a}}_w[k+1])^2 = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\frac{\bar{a}'\,\Delta\hat{\bar{w}}}{\hat{b}}$$

Row 4 of the Jacobian of the exact step (every entry carries the same factor; Section 10 puts it on the diagonal only)

$$\tilde{F}_e(4,4) = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\big(1 + F_{dw}\,\bar{a}'\big)$$

$$\tilde{F}_e(4,3) = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,(1-\lambda_c)\,\bar{a}'$$

$$\tilde{F}_e(4,8) = \tilde{F}_e(4,9) = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\alpha\,\bar{a}'$$

$$\tilde{F}_e(4,5) = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\frac{\bar{a}'\,\Delta\hat{\bar{w}}}{\hat{b}} \qquad \text{(slot 5 free)}$$

$$\tilde{F}_e(4,1) = -\frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,(1-\lambda_c)\,\bar{a}' \qquad \text{(the } n_w \text{ input of Section 9)}$$

$$Q(4,4) = \Big(\frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\Big)^2\,\bar{a}'^2\,Q(3,3), \qquad Q(3,4) = -\frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,\bar{a}'\,Q(3,3)$$

Error dynamics in the $(e_{\delta\bar{w}_3},\ z)$ basis, noise-free, first order (row 3 as in Section 10)

$$\text{exact-step row:} \qquad z[k+1] = \frac{(1-\hat{\bar{a}}_w[k+1])^2}{(1-\hat{\bar{a}}_w[k])^2}\,z[k]$$

$$\text{Euler row (Section 10):} \qquad z[k+1] = \big(1 + A\,\Delta\hat{\bar{w}}\big)\,z[k] + A\,\Delta\hat{\bar{w}}\,\bar{a}'\,\Big((1-\lambda_c)\,\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_3}[k]\big) - F_{dw}\,e_{\bar{a}_w}[k]\Big)$$

(the Euler row is the exact Jacobian of the Euler predict $\hat{\bar{a}}_w + \bar{a}'\Delta\hat{\bar{w}}$ and was kept when the predict became the exact step; per step the two rows differ by $A\Delta\hat{\bar{w}}\times$(coupling), but the difference compounds along the descent as the diagonal does, $\sum A\Delta\hat{\bar{w}} = O(1)$, and the second line forces the law-slaved combination by the position errors that y_1 removes – the $\mathrm{Cov}(z, e_{\delta\bar{w}_1}) \neq 0$ of the third line of this section. Code: ctrl_const.jac_exact_step (default false); acceptance pre-registered in test_script/scratch/run_jac_exact_arms.m; independent sympy verification 09-06: partials, row 4, $z[k+1]$ under both rows and the Q row all PASS – the Euler-row forcing above is the verifier's expression, the first draft had $\lambda_c e_{\delta\bar{w}_3}$ in place of $-(1-\lambda_c)e_{\delta\bar{w}_3}$)

Measured (10 seeds, four blocks, $\kappa = 1$, flag off $\to$ on; canon [Meng]; test_script/scratch/run_jac_exact_arms.m)

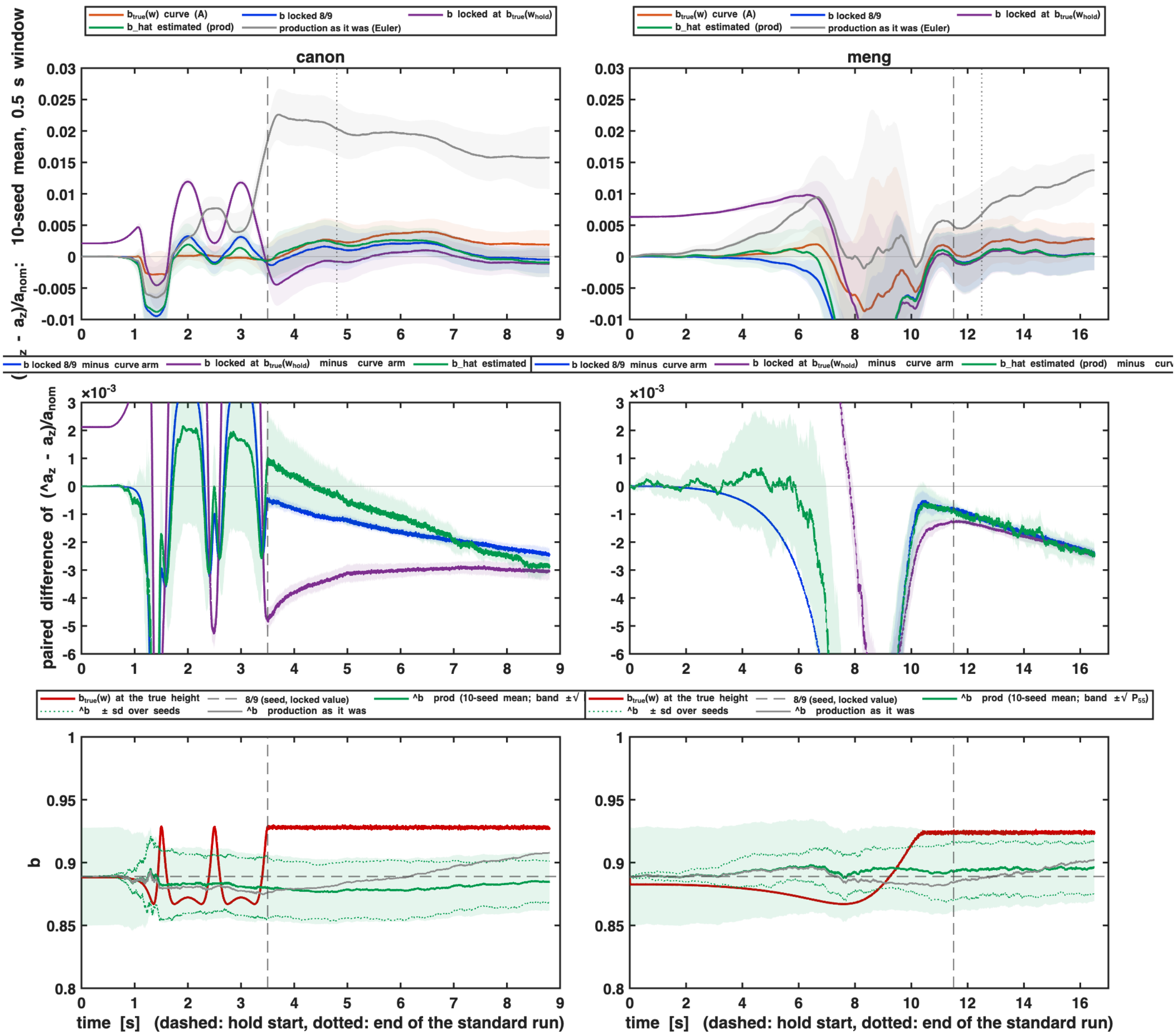
$$\mathrm{b_{true}}\text{ arm:}\quad \sigma_{\text{seed}}\text{ worst instant} : 0.0264 \to 0.0265\ [0.0323 \to 0.0327], \qquad \ell_{41} + \bar{a}'\ell_{31}\text{ (fast)} : +0.1$$
$$\text{hold level, paired on} - \text{off} : +0.00016 \pm 0.00009\ [-0.00006 \pm 0.00006]$$
$$b = 8/9\text{ and }\hat{b}\text{ arms:}\quad \text{paired worst instant} :\leq 0.0002, \qquad \text{paired hold} :\leq 0.00023$$
$$\text{off arms vs ladder\_endpoints (negative control)} : \max|\Delta E| = 0$$

(STATUS: the consistent row is the exact linearisation and is INERT – every paired difference is at the 10^{-4} level on every arm and both trajectories. The row-4 factor is therefore NOT the mechanism of the P_{44} growth or of the spread; the growth is the law's own sensitivity, $\prod(1-\hat{\bar{a}}_w[k+1])^2/(1-\hat{\bar{a}}_w[k])^2 = \bar{a}'_{\text{wall}}/\bar{a}'_{\text{entry}}$, which is the $(1-\bar{a})^2$ growth of a fixed error in $1/(1-\bar{a})$ and is present in truth and model alike. Refuted the same day: $R_2/3$ (K_var/3; the $\mathrm{NIS}_2 = 0.3$ backlog) leaves the fast-window $\sqrt{P_{44}}$ unchanged ($0.0101 \to 0.0100$) and makes the hold over-confident (honesty $0.87 \to 1.65$ canon, $1.08 \to 1.71$ Meng). Flag kept, default false.)

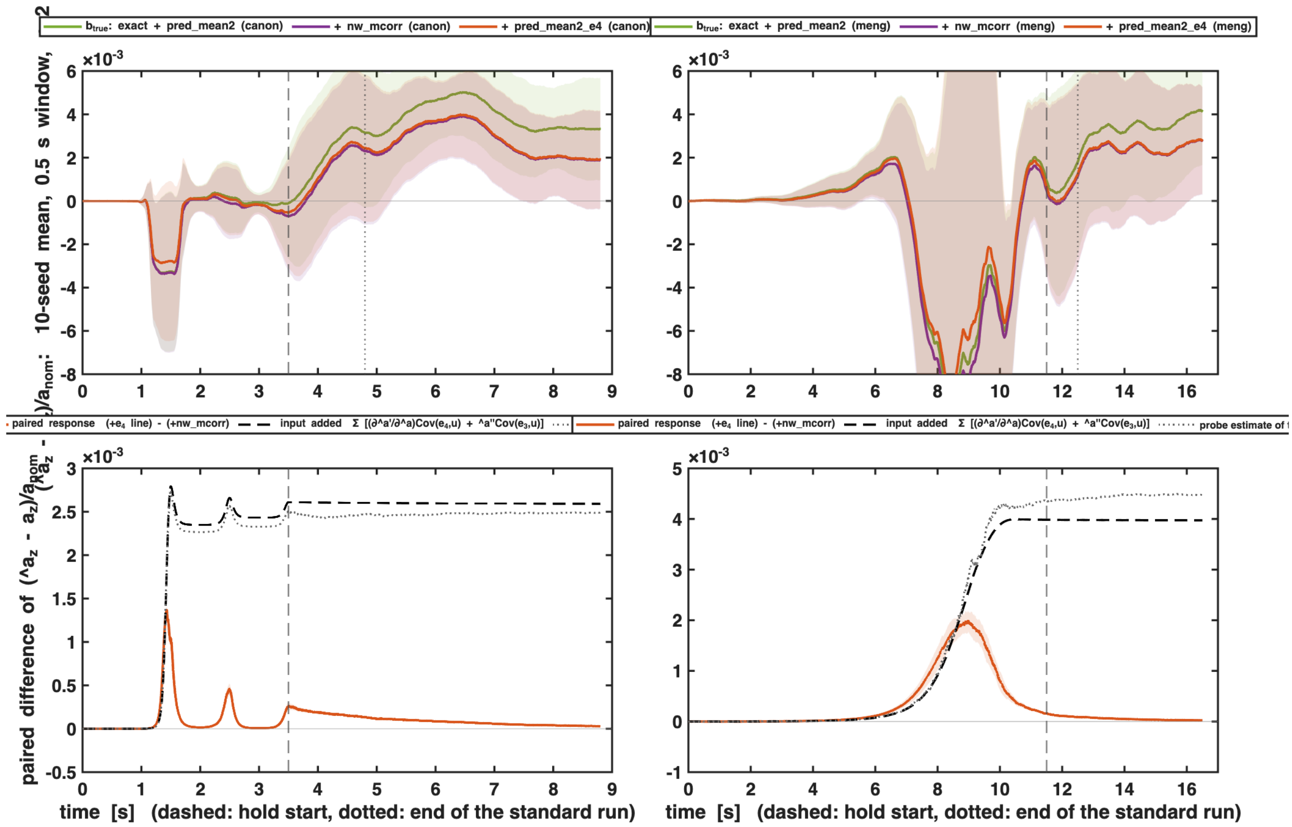
Prior versus a real initial error ($\mathrm{b_{true}}$ arm, 10 seeds, four blocks; test_script/scratch/run_btrue_prior_vs_offset.m

$$P_{44}[0] = \bar{a}'[0]^2\, P_{f,w_0}^2 + P_{f,a,\text{floor}}^2 : \quad \sqrt{P_{44}[0]} = 0.$$
$$\text{seed at truth, } \sqrt{P_{44}[0]} \to 3\times10^{-4}\ [10^{-5}],\ \kappa = 1 : \quad \sigma_{\text{seed}}\text{ worst instant} : 0.0323 \to 0.0051\ [0.0264 \to 0.0035],$$
$$\text{hold: level } +0.0014 \pm 0.0017\ [+0.0010 \pm 0.0010],\ \text{honesty } 1.01\ [1.01], \qquad \text{fast honesty } 0.66\ [0.77]$$
$$\text{seed } +0.0031\text{ (one prior sigma), Meng, } \kappa = 1 : \quad \text{worst-instant mean} = +0.0096, \quad \text{hold } +0.0007 \pm 0.0022, \quad \text{f}$$
$$\text{seed } +0.0031,\ \kappa = 0.5 : \quad \text{worst-instant mean} = +0.0479, \quad \text{hold } +0.0147 \pm 0.0026, \quad \text{f}$$

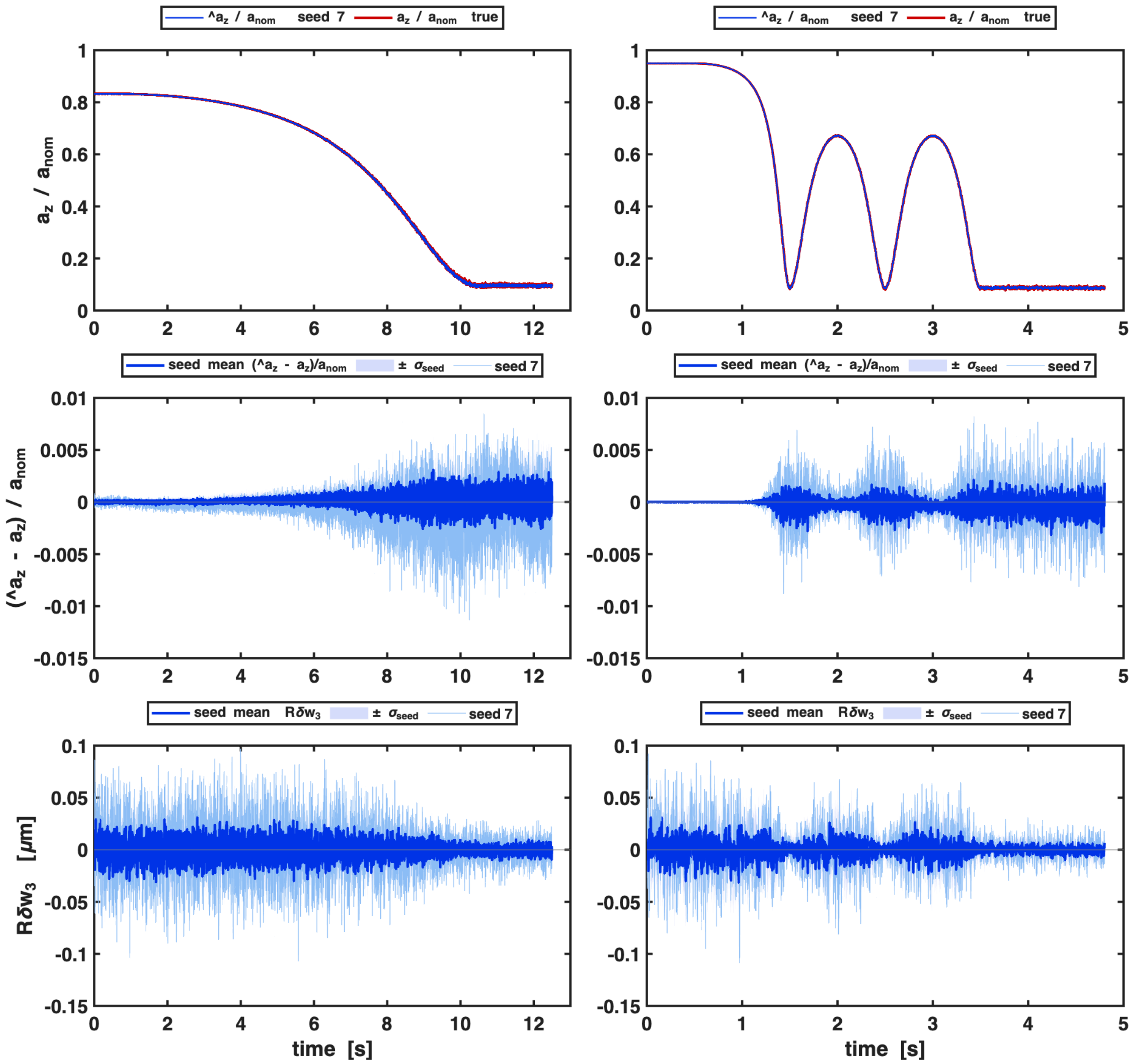
(VERDICT 09-06: the fast-descent P_{44} of the arms that read $\bar{a}'$ at $\hat{\bar{a}}_w$ is the prior $P_{44}[0]$ carried through the law's own sensitivity $\prod(1-\hat{\bar{a}}_w[k+1])^2/(1-\hat{\bar{a}}_w[k])^2$; the realised spread follows it through the gains. In a seed-at-truth simulation that prior is a phantom – matching it to the start removes the spread at $\kappa = 1$ on both trajectories, with the hold honest. With a REAL initial error of one prior sigma, $\kappa = 1$ removes it (hold $+0.0007$) and $\kappa = 0.5$ does not ($+0.0147$, 5.6 SEM; worst instant $+0.048$, the offset amplified $15\times$) – the same failure as the constant-b collapse of the paragraph above. κ is therefore REFUTED as a production quantity in every reading; the near-wall spread is the honest price of the initial gain uncertainty under a self-referential law, and its lever is the seed (calibration chain), not the estimator. The $\mathrm{a}'_{\text{true}}$ arm has no amplification ($A = 0$) and no such spread.)



canon
meng
10-seed mean, 0.5 s window
paired difference of (^a_z - a_z)/a_nom
b_true(w) curve (A)
b_hat estimated (prod)
b locked 8/9
production as it was (Euler)
b locked at b_true(w_hold)
b locked 8/9 minus curve arm
b locked at b_true(w_hold) minus curve arm
b_hat estimated (prod) minus curve arm
b_true(w) at the true height
8/9 (seed, locked value)
^b prod (10-seed mean; band ±√ P55)
^b ± sd over seeds
^b production as it was
b
time [s] (dashed: hold start, dotted: end of the standard run)



b_true: exact + pred_mean2 (canon)
+ nw_mcorr (canon)
+ pred_mean2_e4 (canon)
b_true: exact + pred_mean2 (meng)
+ nw_mcorr (meng)
+ pred_mean2_e4 (meng)
(^a_z - a_z)/a_nom: 10-seed mean, 0.5 s window, 2
paired response (+e4 line) - (+nw_mcorr)
input added Σ [(∂^a'/∂^a)Cov(e4,u) + ^a''Cov(e3,u)]
probe estimate of
paired difference of (^a_z - a_z)/a_nom
time [s] (dashed: hold start, dotted: end of the standard run)
time [s] (dashed: hold start, dotted: end of the standard run)



^a_z / a_nom seed 7
a_z / a_nom true
a_z / a_nom
seed mean (^a_z - a_z)/a_nom
± σ_seed
seed 7
(^a_z - a_z) / a_nom
seed mean Rδw_3
Rδw_3 [μm]
time [s]